REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 10 Lesson 3 | Conservation of Energy

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Ei = Ef for an isolated system; KEi + PEi = KEf + PEf without dissipation; Emech,i = Emech,f + ΔEthermal when appropriate.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 23–32.

P01. In an isolated system, total energy is:

A. Always decreasing B. Always purely kinetic
C. Destroyed by friction D. Conserved

Answer

D. Conserved.

P02. Mechanical energy is the sum of:

A. Kinetic and potential energy B. Mass and speed
C. Force and time D. Momentum and displacement

Answer

A. Kinetic and potential energy.

P03. An elastic collision conserves total:

A. Speed of every object B. Kinetic energy and momentum for an isolated collision
C. Mechanical energy only but not momentum D. Mass times height only

Answer

B. Kinetic energy and momentum for an isolated collision.

P04. In a perfectly inelastic collision, the colliding objects:

A. Always rebound at equal speeds B. Have zero total momentum necessarily
C. Stick together after impact D. Retain all translational kinetic energy

Answer

C. Stick together after impact.

P05. When friction lowers mechanical energy, the missing mechanical energy is mainly:

A. Destroyed B. Converted into negative mass
C. Removed from every possible system D. Transferred to internal or thermal energy

Answer

D. Transferred to internal or thermal energy.

P06. A body starts from rest and descends through height 4 m on a frictionless track. Find its final speed using g = 9.8 m/s².

A. 17.71 m/s B. 4.427 m/s
C. 8.854 m/s D. -8.854 m/s

Answer

C. 8.854 m/s. mgh = ½mv²; v = √(2gh) = √(2 × 9.8 × 4) = 8.854 m/s.

P07. A 2.0 kg block starts from rest and descends 6 m. Friction transfers 10 J of mechanical energy to thermal energy. Find its final kinetic energy; g = 9.8 m/s².

A. 215.2 J B. 53.8 J
C. -107.6 J D. 107.6 J

Answer

D. 107.6 J. KEf = mgh − ΔEthermal = 2.0 × 9.8 × 6 − 10 = 107.6 J.

P08. A ball moves upward at 8 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; g = 9.8 m/s².

A. 3.265 m B. 6.531 m
C. 1.633 m D. -3.265 m

Answer

A. 3.265 m. ½mv² = mgh; h = 8²/(2 × 9.8) = 3.265 m.

P09. A 2 kg cart moving at +6.0 m/s collides with a stationary 4 kg cart. They stick together. Find their common velocity if external impulse is negligible.

A. 4 m/s B. 2 m/s
C. 1 m/s D. -2 m/s

Answer

B. 2 m/s. m1v1 + m2v2 = (m1 + m2)vf; vf = (2 × 6.0 + 4 × 0)/(2 + 4) = +2 m/s.

P10. A 4 kg cart moving +5.0 m/s collides with a 2 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

A. 12 m/s B. 3 m/s
C. 6 m/s D. -6 m/s

Answer

C. 6 m/s. m1v1 + m2v2 = m1v1f + m2v2f; v2f = [4(5.0) + 2(−2.0) − 4(1.0)]/2 = +6 m/s.

P11. State the law of conservation of energy.

Answer

Energy cannot be created or destroyed; it can be transferred or transformed, and total energy remains constant in an isolated system.

P12. When is mechanical energy conserved?

Answer

When the system's energy changes only between kinetic and potential forms and no dissipative or external transfer changes the mechanical total.

P13. Distinguish elastic from inelastic collision.

Answer

An elastic collision conserves total kinetic energy; an inelastic collision does not. Momentum can be conserved in both if external impulse is negligible.

P14. Why should system boundaries be specified in an energy balance?

Answer

They determine which energy transfers cross the boundary and which changes occur internally.

P15. Can a decrease in kinetic energy alone prove total energy was lost?

Answer

No. Energy may have become potential energy, thermal energy, sound or deformation energy, or crossed the chosen boundary.

P16. A body starts from rest and descends through height 6 m on a frictionless track. Find its final speed using $g = 9.8\ m/s^2$.

Answer

$mgh = \frac{1}{2}mv^2$; $v = \sqrt{(2gh)} = \sqrt{(2 \times 9.8 \times 6)} = 10.84$ m/s.

P17. A body starts from rest and descends through height 8 m on a frictionless track. Find its final speed using $g = 9.8\ m/s^2$.

Answer

$mgh = \frac{1}{2}mv^2$; $v = \sqrt{(2gh)} = \sqrt{(2 \times 9.8 \times 8)} = 12.52$ m/s.

P18. A 2.0 kg block starts from rest and descends 9 m. Friction transfers 15 J of mechanical energy to thermal energy. Find its final kinetic energy; $g = 9.8\ m/s^2$.

Answer

$KEf = mgh - \Delta Ethermal = 2.0 \times 9.8 \times 9 - 15 = 161.4$ J.

P19. A 2.0 kg block starts from rest and descends 12 m. Friction transfers 20 J of mechanical energy to thermal energy. Find its final kinetic energy; $g = 9.8\ m/s^2$.

Answer

$KEf = mgh - \Delta Ethermal = 2.0 \times 9.8 \times 12 - 20 = 215.2$ J.

P20. A ball moves upward at 12 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; $g = 9.8\ m/s^2$.

Answer

$\frac{1}{2}mv^2 = mgh$; $h = 12^2/(2 \times 9.8) = 7.347$ m.

P21. A ball moves upward at 16 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; $g = 9.8\ m/s^2$.

Answer

$\frac{1}{2}mv^2 = mgh$; $h = 16^2/(2 \times 9.8) = 13.06$ m.

P22. A 3 kg cart moving at +6.0 m/s collides with a stationary 6 kg cart. They stick together. Find their common velocity if external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; $v_f = (3 \times 6.0 + 6 \times 0)/(3 + 6) = +2$ m/s.

P23. A 4 kg cart moving at +6.0 m/s collides with a stationary 8 kg cart. They stick together. Find their common velocity if external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; $v_f = (4 \times 6.0 + 8 \times 0)/(4 + 8) = +2$ m/s.

P24. A 6 kg cart moving +5.0 m/s collides with a 3 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

Answer

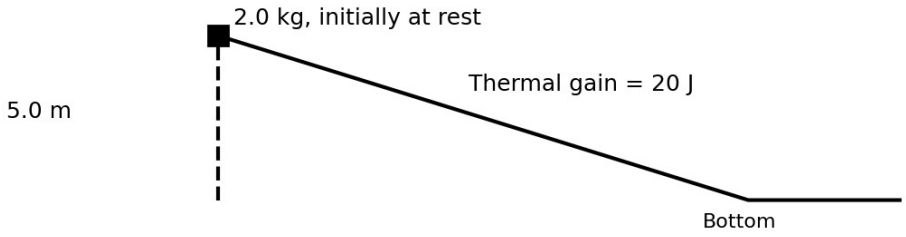
$m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f}$; $v_{2f} = [6(5.0) + 3(-2.0) - 6(1.0)]/3 = +6$ m/s.

P25. A 8 kg cart moving +5.0 m/s collides with a 4 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f}$; $v_{2f} = [8(5.0) + 4(-2.0) - 8(1.0)]/4 = +6$ m/s.

P26. A 2.0 kg block descends 5.0 m from rest and 20 J becomes thermal energy. Find final KE and speed.



Answer

Initial GPE = $2.0 \times 9.8 \times 5.0 = 98$ J. Final KE = $98 - 20 = 78$ J. $v = \sqrt{(2 \times 78/2.0)} \approx 8.83$ m/s.

P27. Two frictionless tracks have different shapes but the same starting and ending heights. An object starts from rest on each. Compare final speeds.

Answer

They are equal because the same vertical drop gives the same decrease in GPE and hence the same final kinetic energy. Path shape does not change the result under the stated assumptions.

P28. Two equal-mass carts collide; one initially moves at 4.0 m/s and one is stationary. If they stick, find final speed and the fraction of initial KE remaining.

Answer

Momentum gives $v_f = 2.0$ m/s. Initial KE = $8m$; final KE = $\frac{1}{2}(2m)(2^2) = 4m$. Half the initial kinetic energy remains.

P29. A student uses mechanical-energy conservation for a rough ramp without a thermal term. Explain the correction.

Answer

Include the increase in thermal/internal energy or work by friction. Mechanical energy alone decreases, while total energy of an enlarged isolated system remains conserved.

P30. Describe an experiment to test approximate mechanical-energy conservation for a pendulum.

Answer

Measure the release height and speed at the lowest point for the same bob, compare mgh with $\frac{1}{2}mv^2$, repeat and account for measurement uncertainty and small frictional losses.